



微积分第三次作业

9(1) $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$

证明: 即对于 $|x \sin \frac{1}{x} - 0| = |x \sin \frac{1}{x}| < \varepsilon$, 找到使其成立的领域 $|x - 0| = |x| < \delta$

$$\because |\sin \frac{1}{x}| \leq 1 \quad \therefore |x \sin \frac{1}{x}| \in |x| < \delta \quad \therefore \text{令 } \delta = \varepsilon \text{ 即可}$$

10) $\lim_{x \rightarrow \infty} \frac{1+2x^2}{2-x^2} = -2$

证明: $\left| \frac{1+2x^2}{2-x^2} - (-2) \right| = \left| \frac{1+2x^2+4x^2-2x^2}{2-x^2} \right| = \left| \frac{5}{2-x^2} \right| < \varepsilon$

(不妨设 $|x| \geq \sqrt{2}$) $\left| \frac{5}{2-x^2} \right| < \frac{5}{x^2-2}$

$$\therefore x^2 > 2 + \frac{5}{\varepsilon}, \quad x > \sqrt{2 + \frac{5}{\varepsilon}} \quad \therefore \text{令 } K = \sqrt{2 + \frac{5}{\varepsilon}} \text{ 即可}$$

11) $\lim_{x \rightarrow 0^+} \ln x = -\infty$

证明: 即对于 $G < 0$, 找到 $x - 0 < \delta$, 使得 $\ln x < G$

$$\therefore x < e^G, \text{ 由指数性质, } e^G > 0, \therefore \text{令 } \delta = e^G \text{ 即可}$$

11. 7) $\lim_{n \rightarrow \infty} \left(\frac{2}{2 \cdot 3 \cdot 4} + \frac{2}{3 \cdot 4 \cdot 5} + \dots + \frac{2}{(n+1)(n+2)(n+3)} \right)$

解: $\frac{2}{2 \cdot 3 \cdot 4} + \frac{2}{3 \cdot 4 \cdot 5} + \dots + \frac{2}{(n+1)(n+2)(n+3)}$ (记为 t_n)

$$= \frac{2 \times 3}{(2 \times 3)(3 \times 4)} + \frac{2 \times 4}{(3 \times 4)(4 \times 5)} + \dots + \frac{2(n+2)}{[(n+1)(n+2)][(n+2)(n+3)]}$$

$$= \frac{1}{2 \times 3} - \frac{1}{3 \times 4} + \frac{1}{3 \times 4} - \frac{1}{4 \times 5} + \dots + \frac{1}{(n+1)(n+2)} - \frac{1}{(n+2)(n+3)}$$

$$= \frac{1}{6} - \frac{1}{(n+2)(n+3)}$$

$$\therefore \lim_{n \rightarrow \infty} t_n = \lim_{n \rightarrow \infty} \left(\frac{1}{6} - \frac{1}{(n+2)(n+3)} \right) = \frac{1}{6} - \lim_{n \rightarrow \infty} \frac{1}{(n+2)(n+3)} = \frac{1}{6} - 0 = \frac{1}{6}$$

(8) $\lim_{n \rightarrow \infty} (1+r)(1+r^2)(1+r^4) \dots (1+r^{2^n}) \quad (|r| < 1)$

解: 记原式为 t_n

$$t_n = (1+r+r^2+r^3)(1+r^4)(1+r^8) \dots (1+r^{2^n})$$

$$= (1+r+r^2+r^3+r^4+r^5+r^6+r^7) \dots (1+r^{2^n}) = \dots$$

$$= 1+r+r^2+r^3+\dots+r^{2^{n+1}-1} = \frac{1-r^{2^{n+1}}}{1-r}$$

$$\lim_{n \rightarrow \infty} t_n = \lim_{n \rightarrow \infty} \frac{1-r^{n+1}}{1-r} = \frac{1 - \lim_{n \rightarrow \infty} r^{n+1}}{1-r} = \frac{1}{1-r}$$

$$(10) \lim_{n \rightarrow \infty} (1 - \frac{1}{2^n}) (1 - \frac{1}{3^n}) \cdots (1 - \frac{1}{n^n})$$

$$\begin{aligned} \text{解: } (1 - \frac{1}{2^n}) (1 - \frac{1}{3^n}) \cdots (1 - \frac{1}{n^n}) &= \frac{1 \times 1}{2^n} \cdot \frac{1 \times 1}{3^n} \cdots \frac{1 \times 1}{n^n} \\ &= \frac{n+1}{2 \times n} = \frac{n+1}{2n} \end{aligned}$$

$$\therefore \lim_{n \rightarrow \infty} (1 - \frac{1}{2^n}) (1 - \frac{1}{3^n}) \cdots (1 - \frac{1}{n^n}) = \lim_{n \rightarrow \infty} (\frac{1}{2} + \frac{1}{2n}) = \frac{1}{2} + \lim_{n \rightarrow \infty} \frac{1}{2n} = \frac{1}{2}$$

$$(11) \lim_{n \rightarrow \infty} n^3 (\frac{k}{n^2} - \sum_{i=1}^k \frac{1}{(n+i)^2}) \quad (k \text{ 为确定的正整数})$$

$$\text{解: } n^3 (\frac{k}{n^2} - \sum_{i=1}^k \frac{1}{(n+i)^2})$$

$$= n^3 [\frac{1}{n^2} - \frac{1}{(n+1)^2} + \frac{1}{n^2} - \frac{1}{(n+2)^2} + \cdots + \frac{1}{n^2} - \frac{1}{(n+k)^2}]$$

$$= n^3 [\frac{1}{n(n+1)} (\frac{1}{n} + \frac{1}{n+1}) + \frac{2}{n(n+2)} (\frac{1}{n} + \frac{1}{n+2}) + \cdots + \frac{k}{n(n+k)} (\frac{1}{n} + \frac{1}{n+k})], \text{ 记为 } t_n$$

$$\lim_{n \rightarrow \infty} t_n = \lim_{n \rightarrow \infty} [\frac{n^2}{n(n+1)} (1 + \frac{n}{n+1}) + \frac{2n^2}{n(n+2)} (1 + \frac{n}{n+2}) + \cdots + \frac{kn^2}{n(n+k)} (1 + \frac{n}{n+k})]$$

$$= (1 \times 2) + (2 \times 2) + \cdots + (k \times 2)$$

$$= 2 \times (1+2+\cdots+k) = k(k+1) = k^2+k$$

$$12. (4) \lim_{n \rightarrow \infty} \sqrt[n]{1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}}$$

$$\text{解: 记 } \sqrt[n]{1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}} = t_n, \quad \frac{1}{n} \text{ 为 } a_n, \quad \frac{1}{n} \text{ 为 } b_n$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 1, \quad \lim_{n \rightarrow \infty} \frac{1}{n} = \lim_{n \rightarrow \infty} e^{\frac{1}{n} \ln n} = 1$$

$$\therefore a_n \leq t_n \leq b_n \quad \therefore \lim_{n \rightarrow \infty} t_n = 1 \quad (\text{由夹逼准则})$$

$$\therefore \lim_{n \rightarrow \infty} \sqrt[n]{1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}} = 1$$

$$13. \text{证明: 设 } M = \max\{a_1, a_2, \dots, a_k\}$$

$$\therefore M^n \leq a_1^n + a_2^n + \cdots + a_k^n \leq M^n + M^n + \cdots = k \cdot M^n$$

$$\therefore M \leq \sqrt[n]{a_1^n + a_2^n + \cdots + a_k^n} \leq \sqrt[k]{k} M$$

$$\therefore \lim_{n \rightarrow \infty} M = M, \quad \lim_{n \rightarrow \infty} \sqrt[k]{k} M = M$$

$$\therefore \lim_{n \rightarrow \infty} \sqrt[n]{a_1^n + a_2^n + \cdots + a_k^n} = M = \max\{a_1, a_2, \dots, a_k\}$$

16. 证明: $x_n = \frac{a_{n+1}}{a_n} = \frac{a_n + a_{n-1}}{a_n} = 1 + \frac{a_{n-1}}{a_n} = 1 + \frac{1}{x_{n-1}}$

$x_1 = \frac{a_2}{a_1} = 1, x_2 = 2, x_3 = \frac{3}{2}, \dots$

下面归纳证明, $1 \leq x_n \leq 2$, 首先对 $n=1$ 成立,

当 $n \leq k$ 成立时, $x_{k+1} = 1 + \frac{1}{x_k} \in [\frac{3}{2}, 2]$, 也成立

$\therefore \{x_n\}$ 有界.

下证 $\{x_n\}$ 从某一项开始递增. 单调递增.

$x_{n+1} - x_n = 1 + \frac{1}{x_n} - x_n = \frac{x_n + 1 - x_n^2}{x_n}$, 对称轴为 $\frac{1}{2}$, 在 $[1, 2]$ 上零点为 $\frac{\sqrt{5}+1}{2}$

当 $x \in [1, \frac{\sqrt{5}+1}{2}]$ 时 (从 x_3 开始), $x_{n+1} - x_n \geq 0$, 递增

当 $x \in [\frac{\sqrt{5}+1}{2}, 2]$ 时, 从 x_3 开始单调增.

$\therefore$ 极限存在. 设为 A , $A \in [1, 2]$

$\therefore A = 1 + \frac{1}{A} \Rightarrow A = \frac{1+\sqrt{5}}{2} \quad \therefore \lim_{n \rightarrow \infty} x_n = \frac{1+\sqrt{5}}{2}$

18 (9) $\lim_{x \rightarrow a} \frac{\cos x - \cos a}{x - a} = \lim_{x \rightarrow a} \frac{-2 \sin \frac{x+a}{2} \sin \frac{x-a}{2}}{x-a} = \lim_{x \rightarrow a} \frac{-(x-a) \sin \frac{x+a}{2}}{x-a}$
 $= \lim_{x \rightarrow a} (-\sin \frac{x+a}{2}) = -\sin a$

19 (iii) $\lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x \cos 3x}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{(1 - \cos x) + \cos x (1 - \cos 2x \cos 3x)}{1 - \cos x}$

~~由例 1.2.15, $1 - \cos x = o(\frac{1}{2}x^2)$, $1 - \cos 2x = o(\frac{1}{2}(2x)^2)$, $1 - \cos 3x = o(\frac{1}{2}(3x)^2)$~~

$= 1 + \lim_{x \rightarrow 0} \frac{\cos x (1 - \cos 2x \cos 3x)}{1 - \cos x} = 1 + \lim_{x \rightarrow 0} \frac{1 - \cos 2x \cos 3x}{1 - \cos x}$

$= \frac{(1 - \cos 2x) + \cos 2x (1 - \cos 3x)}{1 - \cos x} \Big|_{x \rightarrow 0} + 1$

$= 1 + \lim_{x \rightarrow 0} \frac{\cos 2x (1 - \cos 3x)}{1 - \cos x} + \lim_{x \rightarrow 0} \frac{(1 + \cos x) (1 - \cos 2x)}{1 - \cos x} \quad (*)$

由例 1.2.15, $1 - \cos x = o(\frac{1}{2}x^2)$, $1 - \cos 3x = o(\frac{9}{2}x^2)$, $1 - \cos 2x = o(2x^2)$

$(*) = 1 + \lim_{x \rightarrow 0} \frac{1 \times \frac{9}{2}x^2}{\frac{1}{2}x^2} + \lim_{x \rightarrow 0} \frac{2x^2}{\frac{1}{2}x^2} = 14$

20 (13) $\lim_{x \rightarrow 0} \frac{\cos m x - \cos n x}{x^2} = \lim_{x \rightarrow 0} \frac{-2 \sin \frac{m+n}{2} x \cdot \sin \frac{m-n}{2} x}{x^2}$

$= \lim_{x \rightarrow 0} \frac{-2 \cdot \frac{m+n}{2} x \cdot \frac{m-n}{2} x}{x^2} = \frac{n^2 - m^2}{2}$

$$(1b) \lim_{x \rightarrow 0} (1+7x)^{\frac{1}{x}} = \lim_{x \rightarrow 0} e^{\frac{\ln(1+7x)}{x}} = \lim_{x \rightarrow 0} e^{\frac{7x}{x}} = e^7$$

16. 原证明有误。

把原数列分成奇数列和偶数列，奇数列递增，偶数列递减

$\therefore$ 奇数列、偶数列均单调有界，故存在极限。

证明奇数列、偶数列极限相同即可。